Physica A 591 (2022) 126788

Contents lists available at ScienceDirect

Physica A

journal homepage: www.elsevier.com/locate/physa

TSR-GAN: Generative Adversarial Networks for Traffic State Reconstruction with Time Space Diagrams

Kunpeng Zhang [a], Xiaoliang Feng [b], Ning Jia [c], Liang Zhao [a], Zhengbing He [d,*]

[a] College of Electrical Engineering, Henan University of Technology, Zhengzhou, PR China
[b] School of Electrical Engineering, Shanghai Dianji University, Shanghai, PR China
[c] Institute of Systems Engineering, College of Management and Economics, Tianjin University, Tianjin, PR China
[d] Beijing Key Laboratory of Traffic Engineering, Beijing University of Technology, Beijing, PR China

ARTICLE INFO

Article history:
Received 20 March 2021
Received in revised form 17 November 2021
Available online 23 December 2021

Keywords:
Generative Adversarial Networks
Time Space Diagram
Traffic State Reconstruction
Data Imputation

ABSTRACT

A Time Space Diagram (TSD) plays an important role in transportation research and practice due to its capability to exhibit traffic dynamics in time and space. Based on TSDs, this paper aims to reconstruct the traffic spatio-temporal state with the aid of Generative Adversarial Networks (GANs). By mining traffic state correlations and traffic pattern similarities between lanes with or without sufficient observations, the proposed Traffic State Reconstruction GAN (TSR-GAN) model can well estimate the traffic states for road segments with a strong learning capability. Specifically, the traffic states of lanes are converted to TSDs, in which the color represents the values of traffic variables (e.g., speed or density). The TSDs of lanes with or without sufficient data are utilized to train the proposed TSR-GAN model. The fine-tuned TSR-GAN model reconstructs traffic states for road segments with deficient sensor coverage by restoring the high-resolution TSD from its low-resolution observation. With trajectory datasets from Next Generation Simulation (NGSIM), this paper verifies the performance of the TSR-GAN model by estimating travel time via the reconstructed TSDs. Numerical results demonstrate that the proposed model possesses a desirable generalization and transferability, demonstrating the promise of reconstructing traffic states under various conditions.

1. Introduction

Complete and accurate traffic state information plays a vital role in many transportation applications (e.g., operation, control, and congestion mitigation) [1,2]. In practice, however, traffic state variables (e.g., flow, speed, density, travel time) are monitored not everywhere due to technical and financial limitations, probably resulting in the difficulties of supporting traffic control and management [3,4]. To address this problem, numerous studies are dedicated to reconstructing and estimating traffic states to infer traffic variables using partially observed traffic data. These existing methods are divided into three main categories: model-driven approaches, historical data-driven approaches, and streaming data-driven approaches [5].

With the help of prior knowledge and proper assumptions of traffic dynamics, the model-driven approaches are mainly based on physical traffic flow models. In practice, the model-driven approaches are widely used to estimate traffic states under various conditions. Treiber et al. [6,7] proposed the well-known Adaptive Smoothing Method (ASM) and Generalized

* Corresponding author.
E-mail address: he.zb@hotmail.com (Z. He).

https://doi.org/10.1016/j.physa.2021.126788
0378-4371/

Adaptive Smoothing Method (GASM) to interpolate traffic conditions using smoothing kernels. With the consideration of forward and backward wave speed, the adaptive smoothing method was superior to conventional methods without such consideration. With heterogeneous traffic data collected by loop detectors, automatic vehicle identification, and GPS sensors, Deng et al. [8] reconstructed macroscopic freeway traffic states by extending Newell's "three-detector" method. Kessler et al. [9] attempted to reconstruct spatio-temporal traffic speed by discretizing the measured travel times into a Time Space Diagram (TSD) with grid cells to calculate the average speed in the cell. In general, the model-driven approaches are capable of estimating traffic state with a low requirement of massive observations. However, incompetent models, unsatisfied model assumptions, or improper model calibration may lead to undesirable estimation performance.

The historical data-driven approach is empowered by recorded observations instead of traffic flow models. Based on historical observations, the data-driven model employs statistical or machine learning methods to capture the spatio-temporal dependence of traffic state and then carry out the estimation by applying the learned model to target data. To estimate traffic states, many data-driven approaches are proposed in the existing literature. Well-studied methods include autoregressive integrated moving average [10], neural network [11], etc. To handle the greater complexity of traffic data, many studies depend more substantially on past observations, such as machine learning [12,13] and Bayesian statistics [14]. In recent years, various deep learning techniques have been increasingly exploited in traffic state estimation due to its capability of modeling complex dependence within vast amounts of data. With the consideration of spatio-temporal correlations, Lv et al. [15] carried out the traffic flow prediction with a Stacked Autoencoder (SAE) model. With the variants of Recurrent Neural Networks (RNNs) or Convolutional Neural Networks (CNNs), Zhang et al. [16,17] proposed multitask learning frameworks to estimate traffic states by considering spatio-temporal correlations at a network level. Based on CNNs, Benkraouda et al. presented a Convolutional Encoder–Decoder (CED) model to impute traffic data for TSDs [18]. Compared with the model-driven approaches, the historical data-driven approaches do not depend on explicit theoretical assumptions. However, the estimation performance depends on the dependency between historical observations and target data, meaning that the methods may be unqualified if unrecorded events or a long-term trend occurred. Moreover, the computation cost of training the proposed data-driven model can be significantly high, in particular for deep learning-based approaches.

The streaming data-driven approach [19–21] does not strongly rely on prior knowledge and only takes real-time (i.e., streaming) data as input, which enhances its robustness against uncertain events. However, the streaming data-driven approach may claim large sets of streaming data. This means that the accuracy of the streaming data-driven approach may be worse than its counterparts (i.e., model-driven methods and historical data-driven methods) when streaming data is limited. Besides, the streaming data-driven approach cannot predict future traffic states [5].

Recently, many studies attempted to estimate traffic states using TSDs by partitioning a spatio-temporal plane into homogeneous rectangular cells and then estimating traffic states accordingly [22,23]. The TSD is an important visualization tool to identify bottlenecks [24] or understand traffic characteristics [25]. Traditionally, the TSDs are constructed by using stationary detector data or probe vehicle data. However, it is well-known that the stationary detectors are usually installed at a low rate due to their high cost, and the penetration rate of probe vehicles is low as only dedicated vehicles such as taxis can play the role. The shortcomings result in sparsity and insufficiency of the data that can be used to reflect traffic dynamics. In constructing TSDs, large grid cells or data smoothing techniques are necessary to guarantee that most of the cells can be filled with data. The coarse granularity cannot be utilized in high-resolution applications such as estimating travel times and reflecting traffic dynamics.

In this context, the generative model named Generative Adversarial Networks (GANs) [26] draws our attention. As a promising method in the domain of image generation, the GAN is excellent at generating samples by modeling the probability distribution of a dataset [27–30]. Therefore, to reconstruct traffic states by restoring high-resolution TSDs, this paper proposes a Traffic State Reconstruct Generative Adversarial Network (TSR-GAN) model, whose advantages are summarized as follows.

- With low-resolution TSDs from sparse observation, the TSR-GAN model is capable to reconstruct traffic states with desirable generalization, especially when the cell size is large.
- The TSR-GAN model possesses a desirable transferability that can utilize the knowledge learned from road segments in different geographical locations. This indicates that the proposed model can break geographical isolation to capture traffic state similarities when reconstructing traffic states.
- To the best of our knowledge, based on TSDs, the TSR-GAN model may be the first attempt to reconstruct traffic states using GANs. With an advanced model framework and effective loss function, the proposed model possesses reasonable training stability and fine flexibility and demonstrates excellent performance.

The remainder of this paper is organized as follows. Section 2 presents preliminaries and introduces basic concepts. The methodology framework of the TSR-GAN is proposed in Section 3. Section 4 conducts various numerical experiments. Finally, Section 5 draws conclusions.

2. Preliminaries

2.1. Generative Adversarial Networks

The basic idea of GANs is to define a competing game between two networks: the generator network G and the discriminator network D. With a noise vector as input, the generator network G is optimized to generate fake samples.

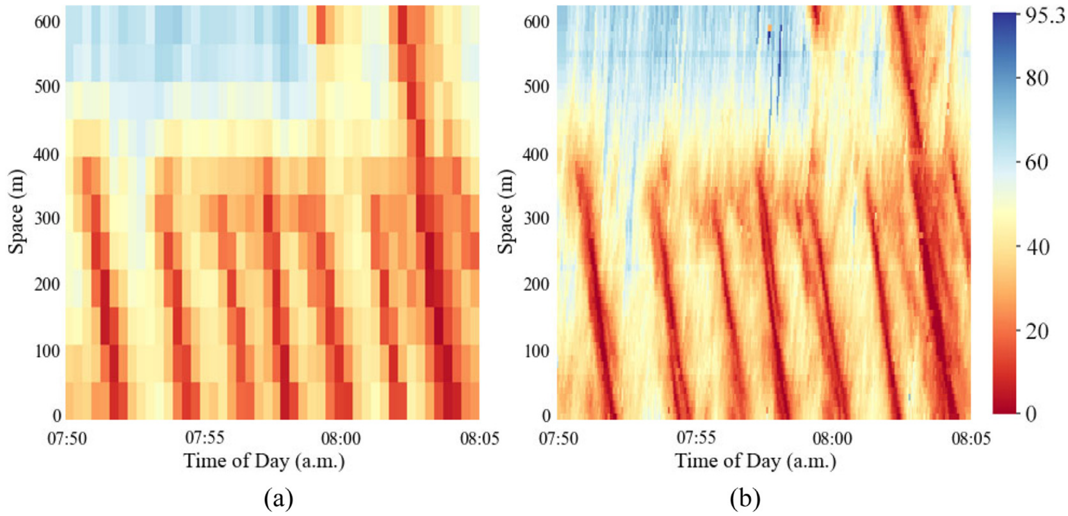

Fig. 1. (a) Low-resolution TSD. (b) High-resolution TSD. *Note:* Color denotes traffic speed (km/h).

With a set of real or generated samples as input, the discriminator network D is trained to improve its skill to differentiate the real samples from the fake ones. The ambition of the generator network G is to make the discriminator network D believe that the fake samples are from the real data distribution. The training procedure of GANs is a min–max game with the objective function as

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{real}} [logD(x)] + \mathbb{E}_{z \sim p_z} [log(1 - D(G(z)))] \tag{1}$$

where x is a real image from the true data distribution p_{real}, and z is a noise vector sampled from a uniform or normal distribution p_z. A fake sample f is from *G(Z)*. The training procedure can be summarized as Algorithm 1.

Algorithm 1 The training procedure of a GAN model

for training iterations **do**
  Initialize hyperparameters of GANs;
  Feed z from p_z into the generator to generate a fake sample f;
  Feed f into the discriminator;
  Feed x from p_{real} into the discriminator;
  Calculate the discriminator loss;
  Maximize *V(D, G)*, update the discriminator;
  Feed z from p_z into the generator to generate f;
  Calculate the generator loss;
  Minimize *V(D, G)*, update the generator.
end for
return the generator

2.2. Time Space Diagram

To construct a TSD, a time space plane is first partitioned into homogeneous rectangular cells with various widths (denoted by w) and heights (denoted by h). Then the traffic speed within each cell can be retrieved from various data sources, such as stationary detector data [31] and probe vehicle data [22]. In a TSD, the x-axis denotes time, the y-axis denotes space, and the color represents traffic speed. Fig. 1(a) illuminates a low-resolution TSD with a large cell size that is caused by sparse traffic data. Accordingly, a high-resolution TSD is shown in Fig. 1(b) with a smaller cell size.

In this paper, the high-fidelity vehicle trajectories from the Next Generation Simulation (NGSIM) program are chosen as the data source. With the help of trajectory data, TSDs are constructed and adopted as the input of the proposed TSR-GAN model.

3. Traffic State Reconstruction Generative Adversarial Networks

In this section, we elaborate on the details of the TSR-GAN model, which is proposed to restore high-resolution TSDs (i.e., with a small cell size) from the low-resolution observations (i.e., with a large cell size). The goal is to recover a high-resolution TSD (i.e., TSD_H) given only a low-resolution TSD (i.e., TSD_L) as input by training the TSR-GAN model to reveal

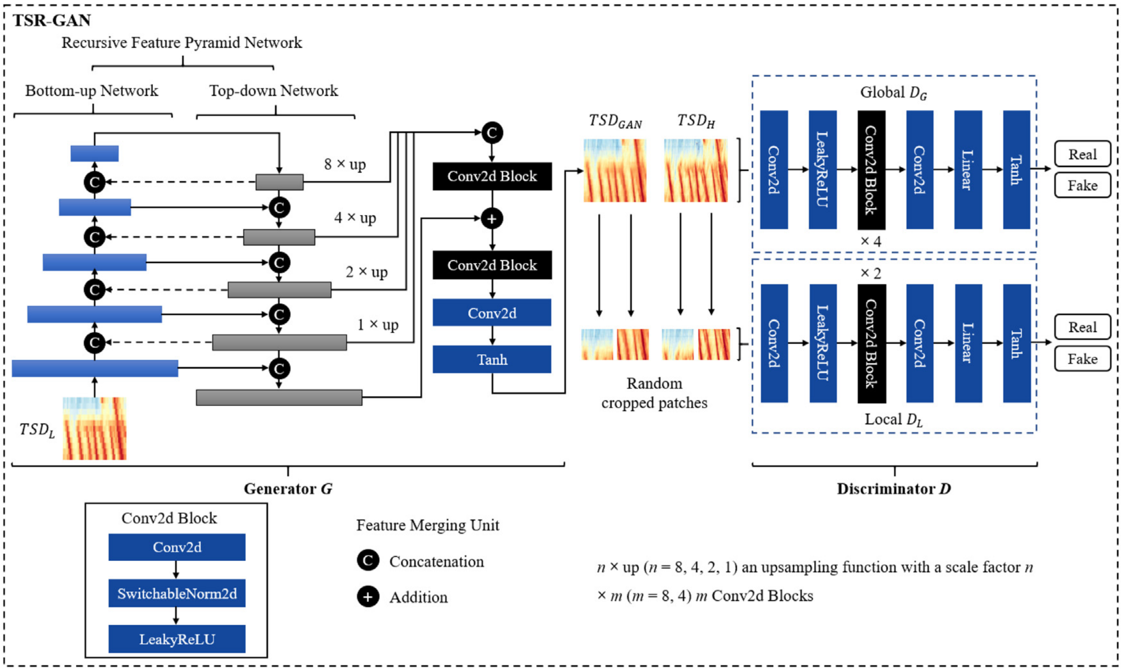

Fig. 2. The Framework of the TSR-GAN.

the fundamental laws that govern traffic dynamics. The TSR-GAN model consists of two components (i.e., the generator network G and the discriminator network D), which are shown in Fig. 2. The generator G and the discriminator D are trained in an adversarial manner. To empower GANs to solve the problems of traffic state reconstruction, the proposed model introduces several novelties in both the model structure and the objective function. Details about the TSR-GAN model and its novelties will be elaborated in the following.

3.1. Generator G with recursive feature pyramid network

In this paper, traffic state reconstruction is achieved by restoring high-resolution TSDs from low-resolution ones with the consideration of traffic state dependence. In the proposed TSR-GAN model, the low-resolution TSD (i.e., TSD_L) with various cell sizes is taken as the input of the generator G. However, it is not a trivial problem to effectively mine useful traffic dynamic information. The reasons can be summarized in two aspects. On one hand, with various cell sizes indicating time and space scales, a series of TSD_L exhibits traffic dynamics in various spatio-temporal dimensions. On the other hand, an TSD_L generally contains a limited number of cells with scarce spatio-temporal traffic information.

To explicitly capture traffic dynamics, a modified Recursive Feature Pyramid Network (RFPN) [32] is introduced to the generator G to extract fundamental traffic state features at a multi-scale level. Stemmed from the Feature Pyramid Networks (FPN) [33] (c.f., Fig. 3(a)), the RFPN (c.f., Fig. 2) consists of a bottom-up network, a top-down network, lateral connections, and look-back connections (i.e., dashed line in Fig. 3(b)). In the TSR-GAN model, with TSD_L as the input, the bottom-up network generally utilizes CNN as the feature extractor, which reduces the sample rate of spatial resolution. However, this practice could extract redundant spatio-temporal information and consequently veil the critical traffic state features. To address this problem, a top-down network is incorporated to enable the RFPN to retrieve higher spatial resolution from information-rich neural network layers. Then, the lateral connections between the bottom-up network and top-down network can capture distinguishable traffic state features. The above-mentioned process can be mathematically described as follows.

Let B_i denote the ith stage of the bottom-up network, and F_i denote the ith top-down RFPN operation. The backbone outputs a set of feature maps $\{f_i|i = 1, \ldots, N\}$, where N is the number of the stages ($N = 3$ in Fig. 3(b)). The output feature f_i can be defined by

$$f_i = F_i\left(f_{i+1}, x_i\right), x_i = B_i(x_{i-1}) \tag{2}$$

where x_0 is the input TSD_L.

To improve the performance, RFPN modifies FPN with look-back connections to transform useful features before connecting them back to the bottom-up network. This modification is inspired by the mechanism of looking and thinking

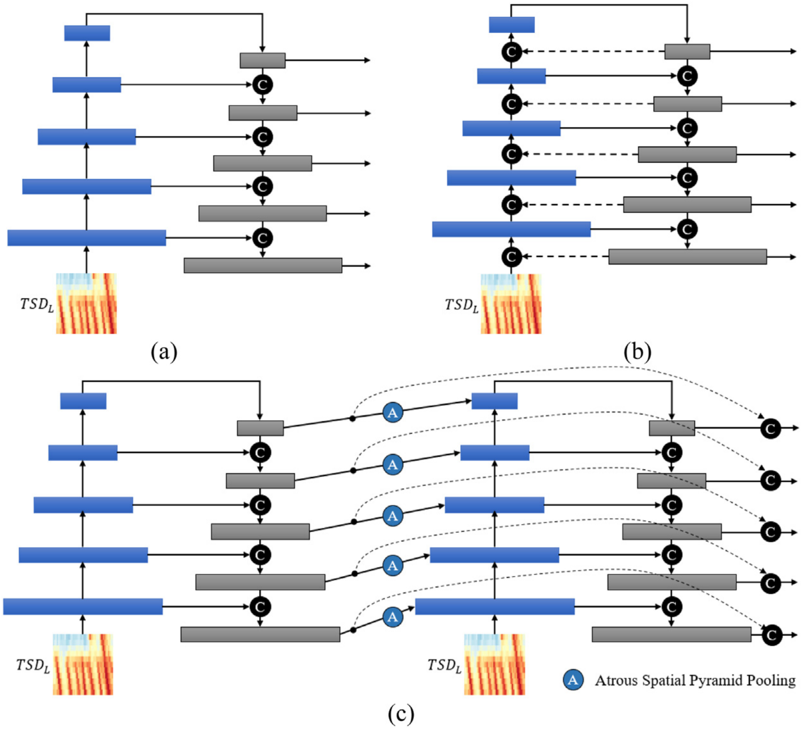

Fig. 3. The structure of (a) Feature Pyramid Network (FPN), (b) Recursive Feature Pyramid Network (RFPN) with look-back connections, and (c) Unrolled RFPN with a 2-step sequential network.

twice in the human visual system. With R_i denotes the feature transformation, RFPN becomes a recursive operation and its output feature f_i can be further defined by

$$f_i = F_i\left(f_{i+1}, x_i\right), x_i = B_i(x_{i-1}, R_i(f_i)) \tag{3}$$

The TSR-GAN with RFPN uses f_i as the traffic dynamic detector. To facilitate the understanding of its computational procedure, the RFPN can be unrolled to a sequential network as shown in Fig. 3(c). The output feature $f_i^t(t = 1, \ldots, T)$ at the unrolled step t can be defined by

$$f_i^t = F_i^t\left(f_{i+1}^t, x_i^t\right), x_i^t = B_i^t(x_{i-1}^t, R_i^t(f_i^{t-1})) \tag{4}$$

To better handle scale variability among TSDs, the TSR-GAN utilizes Atrous Spatial Pyramid Pooling (ASPP) [34] to implement the connection between F_i^t and R_i^t across different steps. The ASPP takes f_i^t as its input and transforms it to the RFPN features to feedback to the corresponding layer in the bottom-up network.

To enhance the learning capacity and training efficiency of the TSR-GAN, the pre-trained Inception-ResNet-v2 [35] is utilized as the backbone in the bottom-up network to extract informative traffic state features. With CNN layers extract feature representations, Inception-ResNet-v2 is capable of effectively producing a multi-scale representation of TSD_L. The max-pooling layers of the Inception-ResNet-v2 render the multi-scale features from CNN layers, which have a pyramid shape. In this way, the bottom levels of features have a higher resolution with less context information. In contrast, the top levels of features have a lower resolution with more context information. In this context, if these features are directly fed into each layer of the top-down network, the outputs from different layers vary significantly, which could cause undesirable outputs due to lacking context information in the lower-level layers.

In order to solve the abovementioned problem, the TSR-GAN proposes a feature pyramid to introduce multi-scale context information to the lower-level layers with the help of the lateral connections and the top-down network. As shown in Fig. 4, the top-down network consists of an Upsample layer, three Conv2d Block, and a ReflectionPad2d layer. With the help of the lateral connections and the top-down network, the TSR-GAN model can reconstruct higher spatial resolution from the context-rich layers. Specifically, the TSR-GAN is capable to detect various scales of traffic state by capturing different extents of traffic features.

The output of each layer in the top-down network is then fed into feature merging units (i.e., Concatenation or Addition in Fig. 2) with upsampling functions to meet the input shape requirement. The outputs of feature merging operation units are fed into the Conv2d Block, Conv2d layer, and Tanh layer (c.f., Fig. 2) to generate the TSD_{GAN}. The generated TSD_{GAN} and the ground truth TSD_H will then be alternatively taken as the input of the discriminator D to improve its skill of distinguishing the generated diagram and the ground truth.

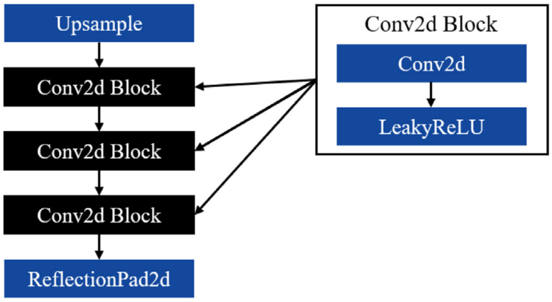

Fig. 4. The structure of the top-down network.

3.2. Discriminator D with bilateral-branch network

In the reconstruction of traffic state, the TSDs generally contain both global and local traffic information. The global traffic features are essential for the discriminator D to incorporate full spatial contexts, while the local traffic features contain more traffic dynamic details. To utilize both global and local features, the TSR-GAN model employs a bilateral-branch network as the discriminator D, consisting of the global branch (i.e., D_G in Fig. 2) that adopts full input TSDs, and the local branch (i.e., D_L in Fig. 2) that operates on patch levels. D_G attempts to capture global traffic state correlations and coarse traffic dynamics, while D_L attempts to capture detailed local traffic dynamics. The TSDs of both the generated TSD_{GAN} and the ground truth TSD_H are randomly cropped into various patches. The random cropped patches are taken as the input of the local discriminator D_L.

3.3. Objective function of TSR-GAN

The objective function mentioned in Eq. (1) is difficult to optimize due to gradient vanishing in the training procedure. To deal with this problem, the discriminator D in the Least Squares GANs (LSGAN) [36] attempts to introduce a loss function that provides a smoother and non-saturating gradient. The study notices that the log-type loss may ignore the distance between the true sample x from p_{real} and the decision boundary, which causes the saturation of the original GAN. In the LSGAN, a least squares loss is introduced to provide proper proportional for the gradients. With the help of the least squares loss, the LSGAN can implement larger penalties for the generated samples farther away from the boundary. The objective function of the LSGAN model is expressed as:

$$\min_D V(D) = \frac{1}{2}\mathbb{E}_{x\sim p_{data}(x)}\left[(D(x)-1)^2\right] + \frac{1}{2}\mathbb{E}_{z\sim p_z(z)}\left[D(G(z))^2\right] \tag{5}$$

$$\min_G V(G) = \frac{1}{2}\mathbb{E}_{z\sim p_z(z)}\left[(D(G(z))-1)^2\right] \tag{6}$$

To solve the problem of gradient vanishing, the Relativistic GAN (RGAN) [37] employs a relativistic discriminator to calculate the probability that real samples are more realistic than random synthetic samples. Compared with its counterparts, the RGAN model performs better in terms of training stability and computational efficiency.

Inspired by the success of LSGAN and RGAN, the TSR-GAN adopts the relativistic wrapping from RGAN on the LSGAN loss function to introduce a new adversarial loss L_{adv}:

$$L_{adv} = \mathbb{E}_{x\sim p_{data}(x)}\left[(D(x) - \mathbb{E}_{z\sim p_z(z)}D(G(Z)) - 1)^2\right] + \mathbb{E}_{z\sim p_z(z)}\left[(D(G(z)) - \mathbb{E}_{x\sim p_{data}(x)}D(x) + 1)^2\right] \tag{7}$$

For training the TSR-GAN model, one needs to compare the reconstructed TSDs and the original ones under certain metrics. With the consideration of representing traffic speed by colors in TSDs, a color loss L_c is introduced in the TSR-GAN model to correct color distortions. This paper chooses the mean square error (MSE) to calculate the color loss L_c, which is formulized as:

$$L_c = \frac{1}{H\times W\times 3}\sum_{i=1}^{H\times W\times 3}(R_i - \hat{R}_i)^2 \tag{8}$$

where R_i and $\hat{R}_i$ are the ith RGB color value of ground truth and the generated TSD with a size of $(H \times W, 3)$.

Given that the cell scale has a direct impact on the representing quality of a TSD for traffic dynamic, the TSR-GAN model employs a scale loss L_s to smoothen pixel-space of the generated TSD_{GAN}. The scale loss L_s is calculated by Kullback–Leibler (KL) divergence, which is formulized as:

$$L_s = KL\left(R \parallel \hat{R}\right) = \sum_{i=1}^{H\times W\times 3} R_i \log(\frac{R_i}{\hat{R}_i}) \tag{9}$$

Therefore, the objective function of the TSR-GAN model is formulized as:

$$L_{TSR-GAN} = \alpha_1 L_{adv} + \alpha_2 L_c + \alpha_3 L_s \tag{10}$$

where $\alpha_i (i = 1, 2, 3)$ denotes the non-zero weight of each component of the $L_{TSR-GAN}$. L_{adv} contains both global and local discriminator losses. After testing various weighted combinations of the objective functions, we find $\alpha_i = 1$ is an appropriate choice.

4. Experiments and results

4.1. Experiment settings

This paper utilizes the high-fidelity vehicle trajectory dataset provided by the NGSIM as the data source. In the NGSIM, the trajectories are collected from the US-101 freeway nearby Lankershim Avenue in Los Angeles, California. On a road segment of 640 m, 6,101 vehicle trajectories were recorded in three consecutive 15-min intervals ranging from 7:35 a.m. to 8:35 a.m. on 15 June 2005. Thus, the spatio-temporal plane is 640 m in height and 2700 s in width. This paper selects the trajectory data collected on Lane 1, 2, 3, 4, and 5. The data provides x, y coordinates and corresponding timestamps of each vehicle. In addition, this paper resamples these trajectories from 0.1 s to 1 s to avoid detection errors [38].

On the basis of Edie's definition [39], if the traffic states of cells are homogeneous, the average traffic variables are qualified to reflect the relations among speed, flow, and density [23]. Therefore, when constructing TSDs, each cell needs to remain in a homogeneous traffic state.

To construct the TSDs, the spatio-temporal plane of US-101 is partitioned into homogeneous cells with various widths and heights. The speeds of all vehicles contained by a cell are extracted from the trajectories and are averaged as the speed of the cell. To train the TSR-GAN, this paper selects the width (w) of each cell from 2 to 300 with an interval of 1 s, and the height (h) from 10 to 300 with an interval of 5 m. After processing the trajectory data, 17,284 TSDs are obtained for each lane. Then, 86,420 TSDs from the five lanes are divided into training, validation, and testing data with division ratios of 70%, 10%, and 20%, respectively.

To measure the quality of the TSR-GAN, the travel time of a vehicle is extracted from the TSDs. Compared to other performance measures (e.g., traffic speed), travel time can provide more straightforward information for travelers to decide route or departure time. Similar to He et al. [23], the trajectory of a vehicle is reconstructed by setting the slope of the trajectory section within a cell to equal the estimated speed of the corresponding cell. Then, the travel time can be calculated with the starting and ending times of the trajectory. The method is simple but widely used [31,40].

To visually evaluate the quality of the TSR-GAN model, the peak signal-to-noise ratio (PSNR) and mean structural similarity (MSSIM) [41] are introduced as traffic state reconstruction quality estimators to compare the ground truth TSD_H and the generated TSD_{GAN}.

The mathematical representation of the PSNR is as follows:

$$PSNR = 20 * log_{10}(\frac{max_r}{\sqrt{MSE}}) \tag{11}$$

$$MSE = \frac{1}{mn} \sum_{0}^{m-1} \sum_{0}^{n-1} \| r\,(i,j) - g(i,j) \|^2 \tag{12}$$

where r represents the matrix data of the ground truth TSD_H; g is the matrix data of the generated TSD_{GAN}; m and n are the numbers of rows and columns of pixels of TSDs, respectively; i and j are the index of rows and columns, respectively; max_r denotes the maximum value that exists in the referred diagram.

The mathematical representation of the MSSIM is as follows:

$$SSIM\left(x_i, y_j\right) = \frac{(2\mu_{x_i}\mu_{y_i} + c_1)(2\sigma_{x_i y_i} + c_2)}{(\mu_{x_i}^2 + \mu_{y_i}^2 + c_1)(\sigma_{x_i}^2 + \sigma_{y_i}^2 + c_2)} \tag{13}$$

$$MSSIM\,(X, Y) = \frac{1}{M} \sum_{j=1}^{M} SSIM(x_i, y_j) \tag{14}$$

where X and Y are the reference diagram (i.e., TSD_H) and the generated diagram (i.e., TSD_{GAN}), respectively. x_j and y_i are the diagram contents at the jth local window, and M is the number of local windows of the diagram. μ_{x_i} and μ_{y_i} denote the mean of x_i and y_i, respectively. $\sigma_{x_i}^2$ and $\sigma_{y_i}^2$ represent the variance of x_i and y_i, respectively. $\sigma_{x_i y_i}$ is the covariance of x_i and y_i. $c_1 = (k_1 L)^2$ and $c_2 = (k_2 L)^2$ is introduced to stabilize the division with a weak denominator. L is the dynamic range of the pixel values. $k_1 = 0.01$ and $k_2 = 0.03$ by default [41].

To be concise and effective, this paper introduces a metric named traffic state similarity (TSS) by referring to PSNR and MSSIM as follows:

$$TSS = (PSNR + MSSIM{*}100)/2 \tag{15}$$

Table 1
Estimation Errors of TSD_H (US-101).

Size ($w \times h$)	RMSE	MAE	MAPE (%)
2×10	2.679	1.598	2.950

To quantitatively evaluate the quality of the TSR-GAN model, this paper retrievals travel time based on the reconstructed TSD. The estimation accuracy is measured by three metrics: root mean squared error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE).

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(Y_i - \hat{Y}_i)^2} \tag{16}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|Y_i - \hat{Y}_i| \tag{17}$$

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\frac{|Y_i - \hat{Y}_i|}{Y_i} \tag{18}$$

where Y_i and $\hat{Y}_i$ are the ith ground truth and the estimated value of travel times, respectively, and N denotes the total number of testing entries.

4.2. Experiment results

To validate the performance of the proposed approaches in estimating travel times with TSD_L, TSD_{GAN} from the TSR-GAN, and TSD_H, the traffic on Lane 1 from 07:50 a.m. to 08:05 a.m. is taken as an example. In this experiment, the TSD with a cell size 2×10 is chosen as the ground truth (i.e., TSD_H). After calculating travel times using TSD_H, three metrics (i.e., RMSE, MAE, and MAPE) are listed in Table 1. Compared with travel times restored from the trajectory data, the TSD_H achieves an acceptable accuracy with RMSE=2.679, MAE=1.598, and MAPE=2.950%. This means the 2×10 diagram is a desirable alternative to represent the real traffic dynamics of the investigated road segment.

To illustrate the performance of the TSR-GAN, this section selects the TSD_L with $w = \{2, 10, 15, 30, 60, 90, 100, 110, 120, 130, 150, 180, 200\}$ s and $h = \{10, 10, 15, 30, 60, 90, 100, 110, 120, 130, 150, 180, 200\}$ m of Lane 1 from 07:50 a.m. to 08:05 a.m. as an example to perform the reconstruction. To ensure fairness, the TSDs of Lane 1 from 07:50 a.m. to 08:05 a.m. are removed from the training dataset. As shown in Fig. 5, the generated TSD_{GAN} is compared with the ground truth TSD_H to validate the effectiveness of the TSR-GAN model with the cell size from 10×10 to 200×200. In general, as the cell size increases, the discrepancy between the generated TSD_{GAN} and the ground truth TSD_H becomes apparent gradually in terms of color, figure resolution, and traffic state features, while the TSS decreases from 43.640 to 29.353. However, compared with the corresponding TSD_L, the estimated TSD_{GAN} is capable to capture more detailed features of the TSD_H even when the cell size reaches 200×200, while the TSS increases from 25.540 to 29.353. In summary, from the point of visualization, with the aid of the TSR-GAN model, the TSD_{GAN} restored from the corresponding TSD_L reveal detailed traffic dynamics and well match with the ground truth TSD_H.

In Table 2, travel times restored from TSDs are detailed to quantify the performance of the TSR-GAN model under different spatio-temporal dimensions. It can be observed that the TSD_{GAN} can remarkably improve the accuracy of travel time estimation in terms of RMSE, MAE, and MAPE. To be specific, compared with the corresponding TSD_L, the TSD_{GAN} is capable to decrease the RMSE, MAE, and MAPE by 18%, 17%, 19% on average. In fact, the benefit of using the TSD_{GAN} is growing as the cell size increases. The reason can be analyzed as follows: Although the traffic dynamics may be erased by the increasing cell sizes, the TSR-GAN model can effectively capture the fundamental traffic features and then restore reasonable TSD_{GAN} from the raw TSD_L.

4.3. Model comparison

To the best of our knowledge, based on TSDs, this paper is the first attempt to reconstruct traffic states using GANs. To validate the effectiveness of the proposed approaches, ten counterpart methods are introduced.

1. GASM: This model employs an asymmetric filter to interpolate missing traffic speed on a TSD by considering information propagation in free and congested traffic [7].
2. CED: Inspired by Benkraouda et al. [18], a CNN-based encoder–decoder framework is introduced to impute missing information for TSDs.
3. SRGAN: Based on GANs, SRGAN is a widely used super-resolution model to estimate a high-resolution image from its low-resolution one [42].

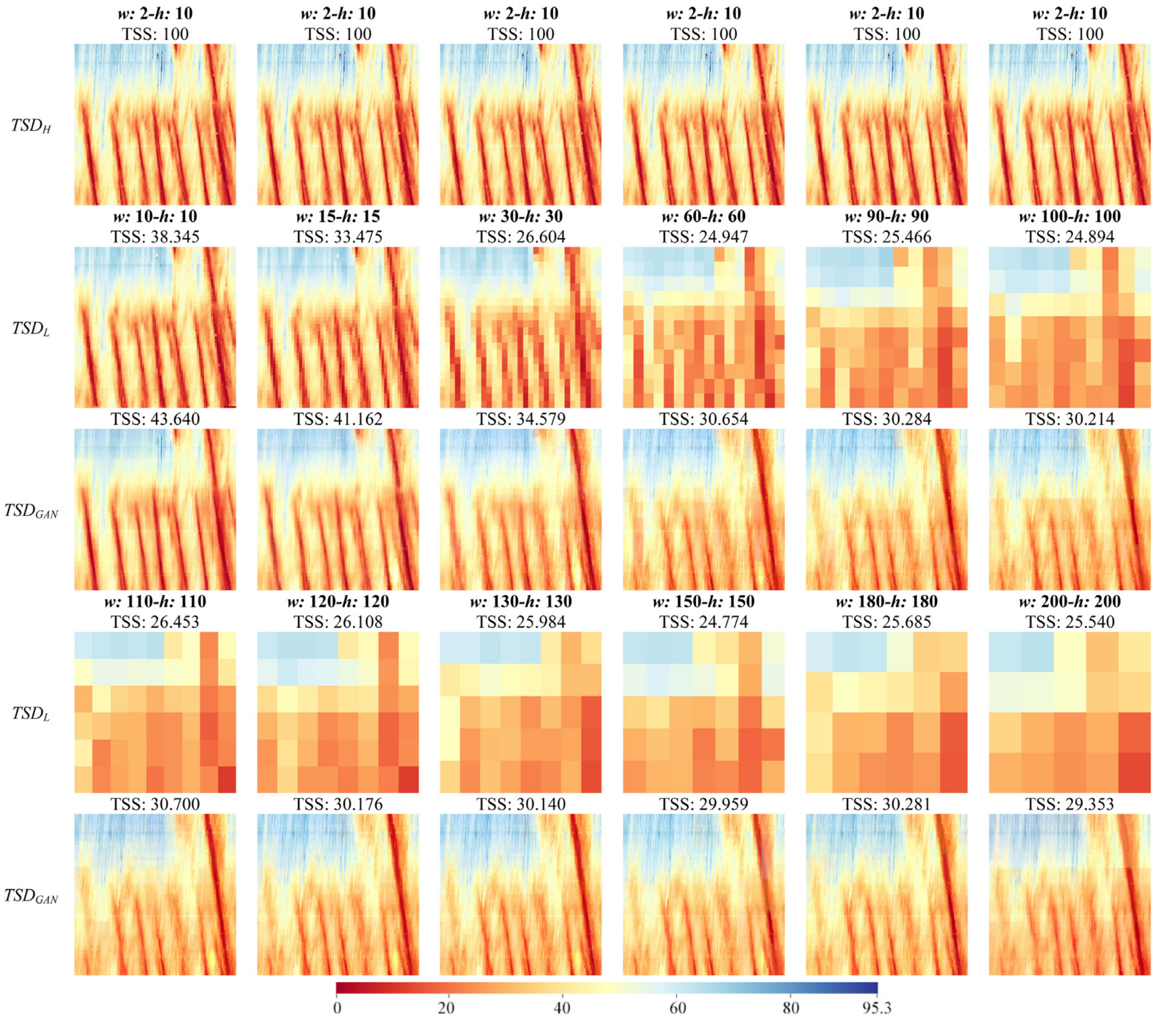

Fig. 5. Comparison of TSD_H, TSD_L, and TSD_{GAN} on Lane1 of US-101. *Note:* Color denotes traffic speed (km/h).

Table 2
Comparison of TSD_L and TSD_{GAN} (US-101).

Size ($w \times h$)	RMSE		MAE		MAPE (%)	
	TSD_L	TSD_{GAN}	TSD_L	TSD_{GAN}	TSD_L	TSD_{GAN}
10 × 10	4.028	**4.022**	2.625	**2.564**	4.490	**4.495**
15 × 15	5.188	**5.189**	3.611	**3.610**	5.759	**5.760**
30 × 30	7.065	**5.694**	5.115	**4.505**	8.031	**7.556**
60 × 60	8.277	**6.279**	6.562	**5.091**	10.901	**8.537**
90 × 90	9.006	**6.153**	7.283	**4.882**	12.337	**8.425**
100 × 100	10.132	**6.404**	8.161	**5.035**	13.990	**8.463**
110 × 110	7.593	**6.875**	6.134	**5.572**	11.030	**9.474**
120 × 120	7.925	**6.983**	6.204	**5.696**	11.102	**9.780**
130 × 130	10.289	**7.493**	8.084	**6.012**	13.905	**9.938**
150 × 150	9.676	**7.611**	7.321	**6.184**	12.299	**10.167**
180 × 180	8.938	**7.918**	7.212	**6.473**	12.272	**10.449**
200 × 200	8.199	**8.394**	6.657	**6.840**	11.606	**11.012**

4. TSR-GAN$_{\mathrm{FPN}}$: This model is proposed by employing FPN in the generator network of the TSR-GAN model instead of RFPN.
5. TSR-GAN$_{\mathrm{local}}$: The discriminator network of TSR-GAN$_{\mathrm{local}}$ only contains the local branch (i.e., D_L in Fig. 2) to mine local traffic information from TSDs.

Table 3
Model Comparison.

Model	TSS	RMSE	MAE	MAPE (%)
TSD_L	27.356	8.026	6.247	10.643
GASM	27.538	8.014	6.236	10.586
CED	28.055	7.931	6.215	10.553
SRGAN	28.591	7.756	6.209	10.543
TSR-GAN$_{FPN}$	29.349	7.850	6.069	10.513
TSR-GAN$_{local}$	30.489	7.460	5.787	9.242
TSR-GAN$_{global}$	30.835	6.943	5.492	8.848
TSR-GAN$_{L-adv}$	29.470	7.689	6.010	10.396
TSR-GAN$_{L-c}$	30.183	7.502	5.827	9.576
TSR-GAN$_{L-s}$	29.968	7.562	5.910	9.587
TSR-GAN$_{SENet}$	32.217	6.633	5.207	8.690
TSR-GAN	**32.595**	**6.585**	**5.205**	**8.671**

6. TSR-GAN$_{global}$: In TSR-GAN$_{global}$, the discriminator network only contains the global branch (i.e., D_G in Fig. 2) to capture global traffic information from TSDs.
7. TSR-GAN$_{L-adv}$: This model has the same model structure as the TSR-GAN model, while the objective function only consists of the adversarial loss L_{adv}.
8. TSR-GAN$_{L-c}$: This model has the same model structure as the TSR-GAN$_{L-adv}$ model, while the objective function consists of L_{adv} and the color loss L_c.
9. TSR-GAN$_{L-s}$: With the same model structure as the TSR-GAN$_{L-c}$ model, this model adopts L_{adv} and the scale loss L_s as its objective function.
10. TSR-GAN$_{SENet}$: TSR-GAN$_{SENet}$ takes the SENet [43] as the backbone of the bottom-up network in the generator network instead of Inception-ResNet-v2 in TSR-GAN.

All experiments are conducted in a desktop computer equipped with Core i7-9700K central processing unit, 32 GB memory, and a GeForce RTX 2080Ti graphics processing unit.

The experimental results are collected and averaged in Table 3. It is found that the TSR-GAN model outperforms the other counterparts in terms of all metrics (i.e., TSS, RMSE, MAE, and MAPE). Specifically, compared with the original TSDs (i.e., TSD_L), the GASM and CED can improve the performance to some degree, while the deep learning-based model (i.e., CED) performs better. Obviously, the SRGAN and TSR-GAN$_{FPN}$ provide a noticeable improvement, which shows the advantage of the GAN-based methods. The results from TSR-GAN$_{local}$ and TSR-GAN$_{global}$ indicate that the global traffic information in TSDs plays a more important role in promoting the performance of the TSR-GAN. Compared with the TSR-GAN$_{L-adv}$ model, the TSR-GAN$_{L-c}$ model and the TSR-GAN$_{L-s}$ model receive better performance. This means that the color loss and the scale loss are valuable to improve the model performance. The TSR-GAN$_{SENet}$ model achieves a comparable performance compared with the TSR-GAN, which shows that the proposed model framework is robust to the changing of the backbone network.

In summary, due to lacking capability of capturing transitions, the GASM is incompetent to reconstruct traffic states, especially when little data is available. Compared with the traditional deep learning-based model (i.e., CED), the GAN-based models are more skillful to capture traffic information due to their refined model structures. With effective objective functions, the TSR-GAN model and its variants perform better than the SRGAN model. Moreover, compared with the FPN, the RFPN can guarantee a better performance when reconstructing traffic states with multi-scale TSDs. The TSR-GAN with the bilateral-branch network as the discriminator network is capable to capture local and global traffic information from TSDs and promises better performance. The color loss and the scale loss play an important role to empower the TSR-GAN to provide qualified traffic state reconstruction. It is also interesting to find that the proposed model framework possesses the potential to be extended with different backbone networks.

4.4. Training stability of TSR-GAN

Take the TSD_L with a cell size 200 × 200 as an example, this section visualizes the generator loss and some intermediate estimated TSD_{GAN} during the training process of the TSR-GAN. The learning procedure and the learning outcome are visualized in Fig. 6. Obviously, the generator loss decreases dramatically during the first training stage and then gradually reduces, which indicates that the TSR-GAN can maintain reasonable stability during the training process. Meanwhile, the intermediate estimated TSD_{GAN} illustrates that the TSR-GAN is gradually optimized to generate better TSD_{GAN} as the training process goes on. In the beginning, with poor skills, the generator G of the TSR-GAN randomly generates a TSD_{GAN} with the TSD_L as input. As the training iteration increase, the skill of the generator G is effectively improved during the competition with the discriminator network, which makes the generated TSD_{GAN} approximate the ground truth TSD_H better and better.

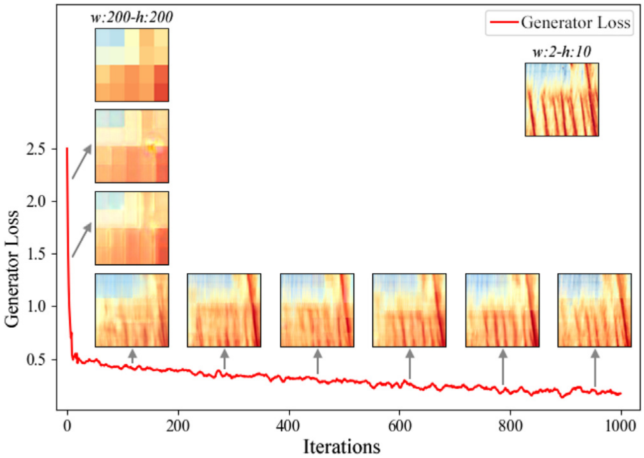

Fig. 6. Visualization of the learning process of TSR-GAN.

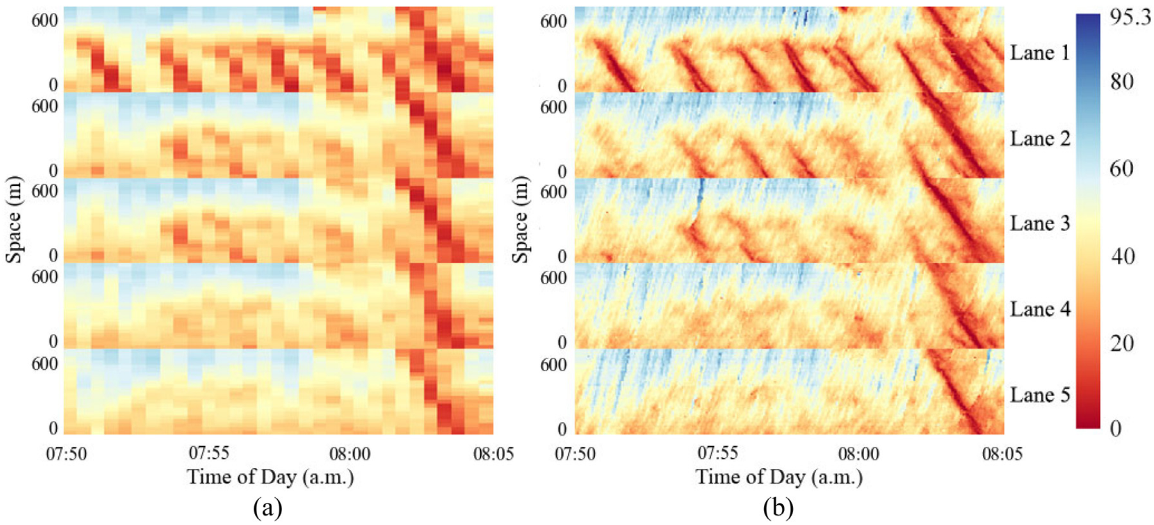

Fig. 7. Unified TSDs. (a) Low-resolution TSD of 5 lanes. (b) High-resolution TSD of 5 lanes.

4.5. Traffic state correlations among neighboring lanes

In practice, the traffic states of neighboring lanes will affect the traffic state reconstruction of a target lane due to traffic state correlations among nearby lanes. To reflect traffic state relations, this section combines TSDs of neighboring lanes to a unified TSD. As shown in Fig. 7, each unified TSD consists of 5 TSDs from 5 lanes (i.e., Lane 1, Lane 2, Lane 3, Lane 4, and Lane 5). With various cell sizes, the unified TSDs with different resolutions are treated as input of the TSR-GAN model to reconstruct the traffic states of the target Lane 1 of the US-101 freeway.

Table 4 details the performance of the TSR-GAN model with TSDs from Lane 1 (i.e., TSD_{GAN-1}) or unified TSDs from 5 lanes (i.e., TSD_{GAN-5}) as input. It can be observed that the TSD_{GAN-5} can improve the accuracy of travel time estimation in terms of RMSE, MAE, and MAPE with the cell size from 10×10 to 200×200. This means the proposed TSR-GAN model is effective to capture traffic state correlations among neighboring lanes to improve the traffic state reconstruction.

4.6. Knowledge transferability

To validate the capability of knowledge transference of the proposed TSR-GAN, this section performs the traffic state reconstruction with the trajectory data of the I-80 highway from NGSIM. This dataset was collected in the San Francisco Bay area in Emeryville, California, on April 13, 2005. A total of 45 min of data are available in the dataset, segmented into three 15-min periods: 4:00 p.m. to 4:15 p.m.; 5:00 p.m. to 5:15 p.m.; and 5:15 p.m. to 5:30 p.m. These periods represent the transition between uncongested and congested conditions and full congestion during the peak period.

Table 4
Performance of TSR-GAN Model with Lane 1 or 5 Lanes (US-101).

Size ($w \times h$)	RMSE		MAE		MAPE (%)	
	TSD_{GAN-1}	TSD_{GAN-5}	TSD_{GAN-1}	TSD_{GAN-5}	TSD_{GAN-1}	TSD_{GAN-5}
10 × 10	4.022	**4.021**	2.564	**2.564**	4.495	**4.490**
15 × 15	5.189	**5.187**	3.610	**3.610**	5.760	**5.742**
30 × 30	5.694	**5.692**	4.505	**4.504**	7.556	**7.537**
60 × 60	6.279	**6.270**	5.091	**5.090**	8.537	**8.516**
90 × 90	6.153	**6.146**	4.882	**4.852**	8.425	**8.406**
100 × 100	6.404	**6.340**	5.035	**5.005**	8.463	**8.439**
110 × 110	6.875	**6.821**	5.572	**5.530**	9.474	**9.460**
120 × 120	6.983	**6.924**	5.696	**5.615**	9.780	**9.751**
130 × 130	7.493	**7.445**	6.012	**6.001**	9.938	**9.903**
150 × 150	7.611	**7.530**	6.184	**6.081**	10.167	**10.072**
180 × 180	7.918	**7.846**	6.473	**6.340**	10.449	**10.356**
200 × 200	8.394	**8.309**	6.840	**6.788**	11.012	**10.903**

Table 5
Estimation Errors of TSD_H (I-80).

Size ($w \times h$)	RMSE	MAE	MAPE (%)
2 × 10	2.164	2.037	3.044

In this experiment, two scenarios are introduced. In the first scenario, with the TSD_L from Lane 6 on I-80 during the third 15-min period, the TSR-GAN model takes only the traffic data from US-101 as the training dataset to generate the corresponding TSD^1_{GAN}. In the second scenario, the TSD_L from the first 15-min period of Lane 2 on I-80 are incorporated in the training dataset to generate the TSD^2_{GAN} for Lane 6 on I-80 during the third 15-min period. With a cell size 2 × 10, the TSD of Lane 6 is chosen as the ground truth (i.e., TSD_H). After calculating travel times using TSD_H, three metrics (i.e., RMSE, MAE, and MAPE) are listed in Table 5. Compared with travel times restored from trajectory data, the TSD_H achieves an acceptable accuracy with RMSE=2.164, MAE=2.037, and MAPE=3.044%. This means the 2 × 10 diagram is a desirable alternative to represent traffic dynamics in the investigated road segment.

The TSS between the estimated TSD_{GAN} and the ground truth TSD_H are dementated Fig. 8(a) to quantify the traffic state similarity between the generated TSD and the ground truth. Obviously, the TSD^2_{GAN} achieves the best TSS. The estimation errors of restored travel times from the TSD_L, TSD^1_{GAN} and TSD^2_{GAN} are demonstrated in Fig. 8(b) (c) (d). It can be observed that the TSD^2_{GAN} achieves the best estimation results. Specifically, with the call size increase from 10 × 10 to 300 × 300, the average RMSEs of TSD_L, TSD^1_{GAN} and TSD^2_{GAN} are 11.133, 10.978, and 9.811, the average MAEs are 8.324, 8.112, and 7.088, and the average MAPEs are 13.469, 13.438, and 12.449, respectively. This means the TSDs from the TSR-GAN model can provide a better estimation. Moreover, when the cell size is larger than 200 × 200, the TSD^1_{GAN} performs worse than the original TSD_L. This can be explained in two aspects. On one hand, as the cell size increases, the original TSD_L bears less instructive information to reflect the traffic dynamic of Lane 6 on I-80. On the other hand, the traffic dynamic characteristics learned only from US-101 may overfit and consequently diminish useful features provided by the TSD_L of Lane 6 on I-80.

In summary, the TSR-GAN model is capable to transfer the knowledge learned from US-101 to reconstruct traffic states of Lane 6 on I-80 by utilizing TSD to represent traffic dynamics. As for TSD^2_{GAN}, the TSS increases compared with TSD^1_{GAN}. This indicates that the TSR-GAN can perform better to construct traffic states of Lane 6 during the third period even with limited traffic information from I-80 during the first period.

5. Conclusions

Based on the widely used TSDs, this paper proposes a TSR-GAN model to reconstruct traffic states. To enhance the learning capability of the proposed model, a generator network with RFPN and a discriminator network with dual branches are introduced with an effective loss function to restore high-resolution TSDs from the low-resolution observations. US-101 and I-80 trajectory datasets provided by NGSIM are employed to test the proposed model. To evaluate the performance of the TSR-GAN, TSS is utilized to qualify the similarity between the reconstructed TSD and the ground truth. With various cell sizes, travel times estimated from the reconstructed TSD are utilized to verify the performance of the TSR-GAN model and ten counterparts. The stability of the TSR-GAN model is analyzed by visualizing its training process. Moreover, the transferability of the TSR-GAN is demonstrated under two scenarios.

The important results and findings are concluded as follows. (1) The TSR-GAN model can reconstruct traffic states for road segments without sufficient traffic data by restoring the high-resolution TSD from its low-resolution observation; (2) The TSR-GAN model possesses a desirable transferability that can utilize the knowledge learned from US-101 to estimate traffic states for I-80; (3) With an advanced model framework and effective loss function, the TSR-GAN model possesses reasonable training stability and fine flexibility. In summary, with a sophisticated model framework, the TSR-GAN model is well-suited to construct traffic states by restoring high-resolution TSDs from the low-resolution ones.

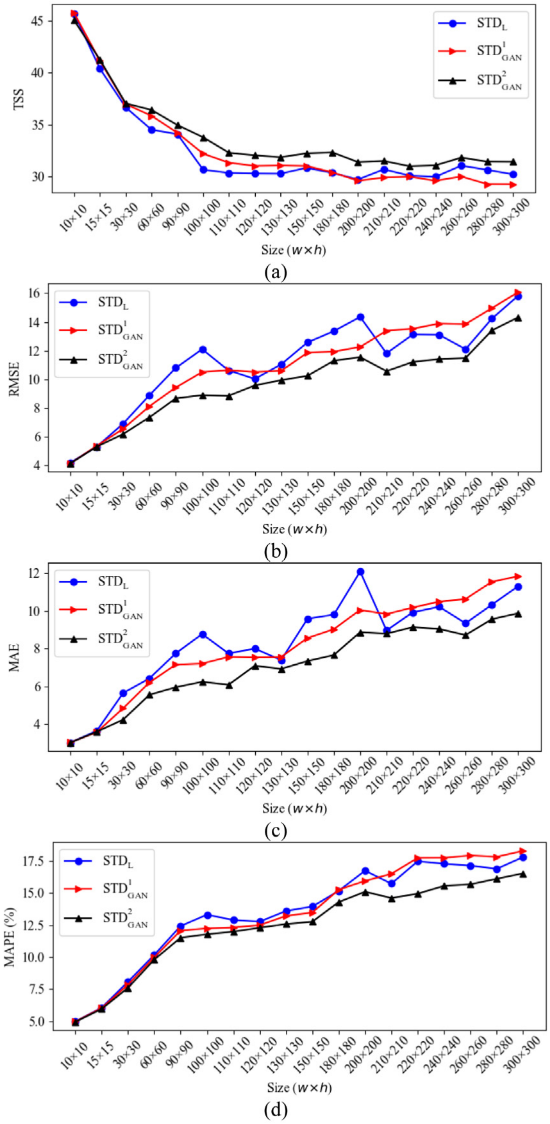

Fig. 8. Comparison of TSD_L, TSD^1_{GAN} and TSD^2_{GAN}. (a) TSS. (b) RMSE. (c) MAE. (d) MAPE.

In the future, a scenario of road segments in an urban area will be investigated and a mixture of heterogeneous data including trajectory data, stationary detector data, and textual data will be utilized to extend the usage of the proposed approaches. In addition, capturing traffic state correlations among nearby lanes deserves to be further explored when reconstructing traffic states.

CRediT authorship contribution statement

Kunpeng Zhang: Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Funding acquisition. **Xiaoliang Feng:** Methodology, Formal analysis, Data curation, Investigation. **Ning Jia:** Methodology, Formal analysis. **Liang Zhao:** Formal analysis, Funding acquisition. **Zhengbing He:** Conceptualization, Methodology, Data curation, Writing – original draft, Writing - review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research has been supported by the National Natural Science Foundation of China (Grants No. 62002101, and 71871010), the Key Scientific Research of Universities in Henan (Grants No. 21A580001), and the Science Foundation of Henan University of Technology (Grants No. 2019BS044).

References

[1] E.I. Vlahogianni, M.G. Karlaftis, J. Golias, Short-term traffic forecasting: Where we are and where we're going, Transp. Res. C 43 (2014) 3–19.
[2] K. Zhang, Z. He, L. Zheng, L. Zhao, L. Wu, A generative adversarial network for travel times imputation using trajectory data, Comput.-Aided Civ. Infrastruct. Eng. 36 (2021) 197–212.
[3] X. Chen, H. Chen, Y. Yang, H. Wu, W. Zhang, J. Zhao, Y. Xiong, Traffic flow prediction by an ensemble framework with data denoising and deep learning model, Physica A 565 (2021) 125574.
[4] J. Tang, J. Hu, W. Hao, X. Chen, Y. Qi, Markov Chains based route travel time estimation considering link spatio-temporal correlation, Physica A 545 (2020) 123759.
[5] T. Seo, A.M. Bayen, T. Kusakabe, Y. Asakura, Traffic state estimation on highway: A comprehensive survey, Annu. Rev. Control 43 (2017) 128–151.
[6] M. Treiber, D. Helbing, Reconstructing the spatio-temporal traffic dynamics from stationary detector data, Cooper. Tran. Dyn. 1 (2002) 3.1–3.21.
[7] M. Treiber, A. Kesting, R.E. Wilson, Reconstructing the traffic state by fusion of heterogeneous data, Comput.-Aided Civ. Infrastruct. Eng. 26 (2011) 408–419.
[8] W. Deng, H. Lei, X. Zhou, Traffic state estimation and uncertainty quantification based on heterogeneous data sources: A three detector approach, Transp. Res. B 57 (2013) 132–157.
[9] L. Kessler, B. Karl, K. Bogenberger, Spatiotemporal traffic speed reconstruction from travel time measurements using bluetooth detection, in: 2019 IEEE Intelligent Transportation Systems Conference, ITSC, IEEE, 2019, pp. 4275–4280.
[10] M. Zhong, P. Lingras, S. Sharma, Estimation of missing traffic counts using factor, genetic, neural, and regression techniques, Transp. Res. C 12 (2004) 139–166.
[11] L. Li, J. Zhang, Y. Wang, B. Ran, Missing value imputation for traffic-related time series data based on a multi-view learning method, IEEE Trans. Intell. Transp. Syst. (2018) 1–11.
[12] S. Tak, S. Woo, H. Yeo, Data-driven imputation method for traffic data in sectional units of road links, IEEE Trans. Intell. Transp. Syst. 17 (2016) 1762–1771.
[13] L. Li, Y. Li, Z. Li, Efficient missing data imputing for traffic flow by considering temporal and spatial dependence, Transp. Res. C 34 (2013) 108–120.
[14] K. Tang, S. Chen, Z. Liu, A.J. Khattak, A tensor-based Bayesian probabilistic model for citywide personalized travel time estimation, Transp. Res. C 90 (2018) 260–280.
[15] Y. Lv, Y. Duan, W. Kang, Z. Li, F.-Y. Wang, Traffic flow prediction with big data: a deep learning approach, IEEE Trans. Intell. Transp. Syst. 16 (2014) 865–873.
[16] K. Zhang, L. Zheng, Z. Liu, N. Jia, A deep learning based multitask model for network-wide traffic speed prediction, Neurocomputing 396 (2019) 438–450.
[17] K. Zhang, Z. Liu, L. Zheng, Short-term prediction of passenger demand in multi-zone level: Temporal convolutional neural network with multi-task learning, IEEE Trans. Intell. Transp. Syst. 21 (2019) 1480–1490.
[18] O. Benkraouda, B.T. Thodi, H. Yeo, M. Menendez, S.E. Jabari, Traffic data imputation using deep convolutional neural networks, IEEE Access 8 (2020) 104740-104752.
[19] T. Seo, T. Kusakabe, Y. Asakura, Estimation of flow and density using probe vehicles with spacing measurement equipment, Transp. Res. C 53 (2015) 134–150.
[20] T.Z. Qiu, X.-Y. Lu, A.H. Chow, S.E. Shladover, Estimation of freeway traffic density with loop detector and probe vehicle data, Transp. Res. Rec. 2178 (2010) 21–29.
[21] N. Bekiaris-Liberis, C. Roncoli, M. Papageorgiou, Highway traffic state estimation with mixed connected and conventional vehicles, IEEE Trans. Intell. Transp. Syst. 17 (2016) 3484–3497.
[22] Z. He, L. Zheng, P. Chen, W. Guan, Mapping to cells: A simple method to extract traffic dynamics from probe vehicle data, Comput.-Aided Civ. Infrastruct. Eng. 32 (2017) 252–267.
[23] Z. He, Y. Lv, L. Lu, W. Guan, Constructing spatiotemporal speed contour diagrams: using rectangular or non-rectangular parallelogram cells? Transportmetr. B: Transp. Dyn. 7 (2019) 44–60.
[24] X. Ban, L. Chu, H. Benouar, Bottleneck identification and calibration for corridor management planning, Transp. Res. Rec. 1999 (2007) 40–53.
[25] Z. He, S. He, W. Guan, A figure-eight hysteresis pattern in macroscopic fundamental diagrams and its microscopic causes, Transp. Lett. 7 (2015) 133–142.
[26] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, Y. Bengio, Generative adversarial nets, in: Advances in Neural Information Processing Systems, 2014, pp. 2672–2680.
[27] Y. Chen, Y. Lv, F.-Y. Wang, Traffic flow imputation using parallel data and generative adversarial networks, IEEE Trans. Intell. Transp. Syst. (2019).
[28] Y. Lin, X. Dai, L. Li, F.-Y. Wang, Pattern sensitive prediction of traffic flow based on generative adversarial framework, IEEE Trans. Intell. Transp. Syst. 20 (2018) 2395–2400.
[29] Y. Lv, Y. Chen, L. Li, F.-Y. Wang, Generative adversarial networks for parallel transportation systems, IEEE Intell. Transp. Syst. Mag. 10 (2018) 4–10.
[30] K. Zhang, N. Jia, L. Zheng, Z. Liu, A novel generative adversarial network for estimation of trip travel time distribution with trajectory data, Transp. Res. C 108 (2019) 223–244.
[31] M. Yildirimoglu, N. Geroliminis, Experienced travel time prediction for congested freeways, Transp. Res. B 53 (2013) 45–63.
[32] S. Qiao, L.-C. Chen, A. Yuille, DetectoRS: detecting objects with recursive feature pyramid and switchable atrous convolution, 2020, arXiv preprint arXiv:2006.02334.
[33] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, S. Belongie, Feature pyramid networks for object detection, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 2117–2125.
[34] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, A.L. Yuille, Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs, IEEE Trans. Pattern Anal. Mach. Intell. 40 (2017) 834–848.
[35] C. Szegedy, S. Ioffe, V. Vanhoucke, A.A. Alemi, Inception-v4, inception-resnet and the impact of residual connections on learning, in: Thirty-first AAAI Conference on Artificial Intelligence, 2017.

[36] X. Mao, Q. Li, H. Xie, R.Y. Lau, Z. Wang, S. Paul Smolley, Least squares generative adversarial networks, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2794–2802.
[37] A. Jolicoeur-Martineau, The relativistic discriminator: A key element missing from standard GAN, 2018, arXiv preprint arXiv:1807.00734.
[38] V. Punzo, M.T. Borzacchiello, B. Ciuffo, On the assessment of vehicle trajectory data accuracy and application to the Next Generation SIMulation (NGSIM) program data, Transp. Res. C 19 (2011) 1243–1262.
[39] L. Edie, Discussion of traffic stream measurements and definitions, in: 2nd Int. Symp. on the Theory of Traffic Flow, OECD, Paris, 1965.
[40] M. Elhenawy, H. Chen, H.A. Rakha, Dynamic travel time prediction using data clustering and genetic programming, Transp. Res. C 42 (2014) 82–98.
[41] Z. Wang, A.C. Bovik, H.R. Sheikh, E.P. Simoncelli, Image quality assessment: From error visibility to structural similarity, IEEE Trans. Image Process. 13 (2004) 600–612.
[42] C. Ledig, L. Theis, F. Huszár, J. Caballero, A. Cunningham, A. Acosta, A. Aitken, A. Tejani, J. Totz, Z. Wang, Photo-realistic single image super-resolution using a generative adversarial network, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2017, pp. 4681–4690.
[43] J. Hu, L. Shen, G. Sun, Squeeze-and-excitation networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 7132–7141.